

Untitled

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2025-04-13

R Markdown

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.2
```

```
# === 1. Load Data ===
```

```
data_raw <- read.table("interexp.dat", header = FALSE, stringsAsFactors = FALSE)
colnames(data_raw) <- c("stimA", "stimB")
```

```
data <- data.frame(
  stimA = as.numeric(gsub("[^0-9\\.]", "", data_raw$stimA)),
  stimB = as.numeric(gsub("[^0-9\\.]", "", data_raw$stimB))
)
```

```
data <- data[!(is.na(data$stimA) & is.na(data$stimB)), ]
```

```
YA <- data$stimA
YB <- data$stimB
```

```
N <- nrow(data)
```

```
# === 2. Initialization ===
```

```
muA <- mean(YA, na.rm = TRUE)
muB <- mean(YB, na.rm = TRUE)
sigmaA2 <- var(YA, na.rm = TRUE)
sigmaB2 <- var(YB, na.rm = TRUE)
rho <- cor(YA, YB, use = "complete.obs")
```

```
tol <- 1e-6
max_iter <- 100
converged <- FALSE
```

```
EYA <- YA
EYB <- YB
```

```
# === 3. EM Algorithm ===
```

```
for (iter in 1:max_iter) {
  idx_A_missing <- is.na(YA) & !is.na(YB)
  idx_B_missing <- !is.na(YA) & is.na(YB)

  EYA[idx_A_missing] <- muA + rho * sqrt(sigmaA2) / sqrt(sigmaB2) * (YB[idx_A_missing] - muB)
```

```

EYB[idx_B_missing] <- muB + rho * sqrt(sigmaB2) / sqrt(sigmaA2) * (YA[idx_B_missing] - muA)

muA_new <- mean(EYA)
muB_new <- mean(EYB)
sigmaA2_new <- mean(EYA^2) - muA_new^2
sigmaB2_new <- mean(EYB^2) - muB_new^2
covAB <- mean(EYA * EYB) - muA_new * muB_new
rho_new <- covAB / sqrt(sigmaA2_new * sigmaB2_new)

diffs <- c(abs(muA_new - muA), abs(muB_new - muB),
           abs(sigmaA2_new - sigmaA2), abs(sigmaB2_new - sigmaB2),
           abs(rho_new - rho))
if (max(diffs) < tol) {
  converged <- TRUE
  break
}

muA <- muA_new; muB <- muB_new
sigmaA2 <- sigmaA2_new; sigmaB2 <- sigmaB2_new; rho <- rho_new
}

cat("Converged in", iter, "iterations\n")

## Converged in 14 iterations
cat(sprintf("muA = %.3f, muB = %.3f\n", muA, muB))

## muA = 24.212, muB = 24.830
cat(sprintf("varA = %.3f, varB = %.3f, rho = %.3f\n", sigmaA2, sigmaB2, rho))

## varA = 3.515, varB = 3.711, rho = 0.750
# === 4. Paired T-Tests ===

complete_idx <- complete.cases(YA, YB)
pre_ttest <- t.test(YA[complete_idx], YB[complete_idx], paired = TRUE)
print(pre_ttest)

##
## Paired t-test
##
## data: YA[complete_idx] and YB[complete_idx]
## t = -1.6051, df = 25, p-value = 0.121
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -1.4216704 0.1762858
## sample estimates:
## mean difference
## -0.6226923

post_ttest <- t.test(EYA, EYB, paired = TRUE)
print(post_ttest)

##
## Paired t-test
##

```

```
## data: EYA and EYB
## t = -3.4662, df = 57, p-value = 0.00101
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -0.9742842 -0.2607700
## sample estimates:
## mean difference
## -0.6175271

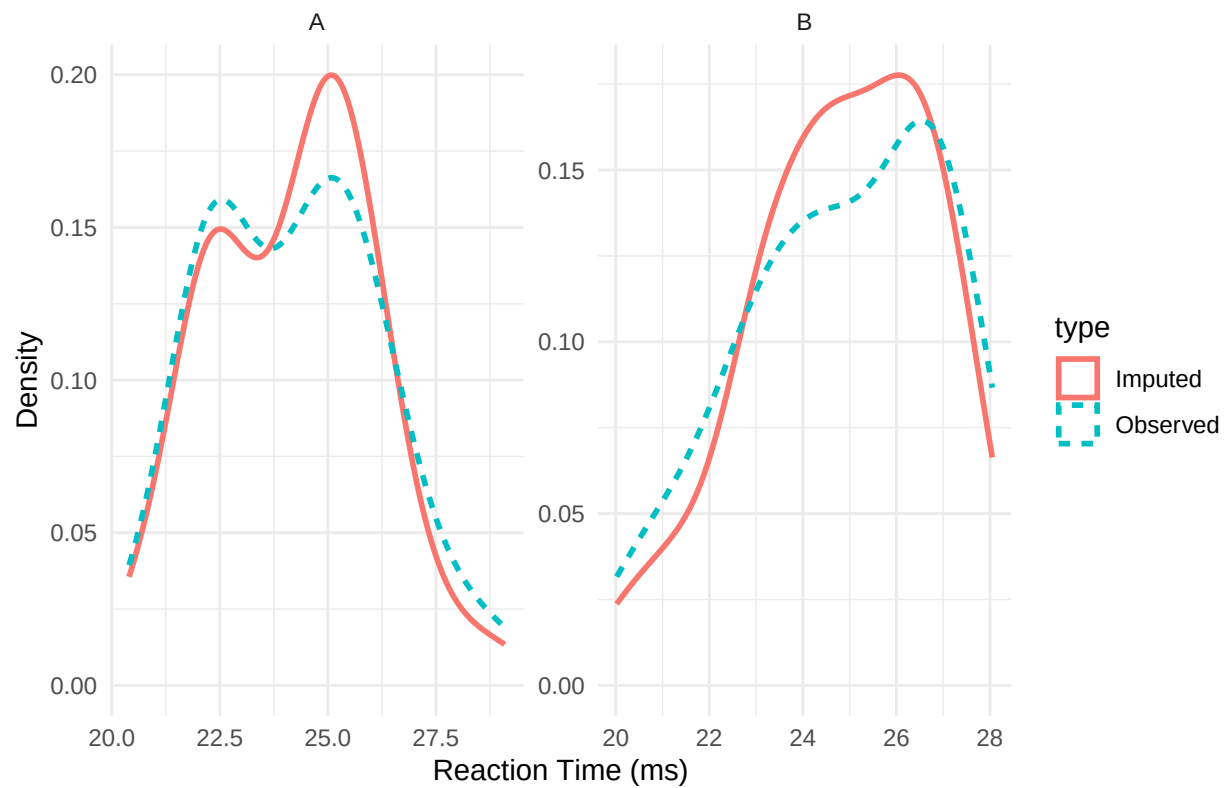
# === 5. Plot Distributions ===
df_long <- data.frame(
  value = c(YA, EYA, YB, EYB),
  stimulus = rep(c("A", "A", "B", "B"), each = N),
  type = c(rep("Observed", N), rep("Imputed", N), rep("Observed", N), rep("Imputed", N))
)

ggplot(df_long, aes(x = value, color = type, linetype = type)) +
  geom_density(size = 1) +
  facet_wrap(~stimulus, scales = "free") +
  labs(title = "Reaction Time Distributions (Stimulus A vs B)",
       x = "Reaction Time (ms)", y = "Density") +
  theme_minimal()

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Warning: Removed 32 rows containing non-finite outside the scale range
## (`stat_density()`).
```

Reaction Time Distributions (Stimulus A vs B)



```
# === 6. Export Imputed Dataset ===  
completed_data <- data.frame(StimulusA = EYA, StimulusB = EYB)  
write.csv(completed_data, "new_dataset.csv", row.names = FALSE)
```